



Web of Statistics

Turning Data into Knowledge

User Manual

www.webofstatistics.com

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1. About Web of Statistics.

Web of Statistics is a free online platform that provides statistical analysis, machine learning, data visualization, and consumer sensory analysis tools. Its available resources include descriptive statistics, probability functions and distribution tables, statistical inference, statistical process control charts, analysis of variance and multivariate analysis of variance, regression, classification methods, tree-based models, principal component analysis, clustering, multivariate regression, and sensory evaluation methods.

The platform was created and developed by Dr. Silvio D. Rodríguez, Professor of Applied Statistics at the Pontificia Universidad Católica Argentina (UCA), Buenos Aires. It was developed using the Django web framework with the aim of making statistical tools more accessible to students, teachers, researchers, and professionals.

Web of Statistics provides calculators, graphical tools, and predictive models that can be used directly from a web browser, without requiring users to install additional software. Its main purpose is to support education, research, and professional data analysis by offering accessible resources for learning, applying, and interpreting statistical methods.

The platform is an academic and educational initiative. It is also an open-source project licensed under the GNU General Public License v3. Its source code is publicly available through the project's GitHub repository, allowing users and developers to inspect, study, contribute to, and improve the software.

The statistical tools are provided free of charge. However, users are responsible for selecting appropriate methods, verifying the accuracy and suitability of the input data, interpreting the results correctly, and determining whether the analyses are appropriate for their intended purpose. The author assumes no responsibility for errors, misinterpretations, decisions, or consequences resulting from the use of the platform or any of its components.

Users are invited to consider making a donation to help cover hosting and maintenance costs. Contributions help ensure that the platform remains freely available and free from advertising.

Feedback, error reports, and suggestions for improvement are welcome. These contributions are valuable for improving the platform, correcting potential issues, and ensuring that its tools continue to meet users' needs. Comments and suggestions may be sent to webofstatistics@gmail.com

Accessing the Platform

To access the statistical tools, users should visit www.webofstatistics.com. The home page provides general information about the platform, its purpose, and its author. It also includes links to the user documentation, example datasets, the project's GitHub repository, scientific publications in which Web of Statistics has been used, and the Community Forum.

The Example Datasets link opens a shared Google Sheets document containing datasets organized into separate worksheets. These examples are designed to help users understand the required data structure and test the tools available in the different sections of the platform.

The Scientific Publications section presents articles and research projects in which Web of Statistics has been used for data analysis.

The Community Forum allows users to ask questions, share experiences, and discuss statistical methods and the use of the platform. Topics and replies are reviewed by an administrator before publication.



The sidebar provides access to the main sections of the platform. Each statistical module and its corresponding tools are described in the following chapters of this user manual.

2. Descriptive Statistics.

Clicking the 'Descriptive' tab opens the Descriptive Statistics section in the main content area. A brief description of the available tools appears at the top of the page, followed by a navigation menu. By default, the 'Bulk Data' option is selected. This section provides basic descriptive statistical calculations for tabulated datasets. The three available options are 'Bulk Data,' 'Histograms,' and 'Boxplot.'

Descriptive Statistics Calculator
Summary Statistics, Histograms and Boxplots
Use this free descriptive statistics calculator to compute summary statistics, generate histograms and create boxplots for exploratory data analysis and distribution assessment.

Bulk Data | Histograms | Boxplot

Bulk Data
Data
Paste data from Excel or Google Sheets (each variable in a different column).
First Row: Variable Labels (optional).

☐ Use first row as variable names

Calculate **Clean Data & Results**

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webotstatistics@gmail.com

After selecting an option from the top menu, enter the data in the data input area. To make the platform easier to use, the data entry procedure is consistent across its sections and subsections. Follow the instructions for organizing the data, then copy and paste the values from a spreadsheet.

For example, when 'Bulk Data' is selected, each variable should be placed in a separate column. The first row may optionally contain the variable names. When names are included, select the 'Use first row as variable names' checkbox. When the first row contains only data, leave the checkbox unselected; the platform will automatically assign a name to each variable.

The following example shows data copied from a spreadsheet. It contains three variables, although a dataset may contain more or fewer columns. Copy the selected cells and paste them into the data input area by using the mouse or the keyboard shortcuts Ctrl+C and Ctrl+V.

A1 Riboflavin (ppm)	A2 Riboflavin (ppm)	A3 Riboflavin (ppm)
0.169	0.187	0.157
0.170	0.174	0.121
0.130	0.192	0.127
0.168	0.103	0.109
0.183	0.119	0.166



0.160	0.117	0.173
0.167	0.171	0.190
0.166	0.192	0.148
0.139	0.175	0.130
0.199	0.125	0.135
0.110	0.131	0.148
0.146	0.143	0.172
0.183	0.152	0.185
0.167	0.190	0.106
0.135	0.176	0.118
0.148	0.160	0.199

The pasted data will appear on the page as shown below. Use a period (.) as the decimal separator. Values may also be entered in scientific notation.

Bulk Data
Histograms
Boxplot

Bulk Data

Data

A1 Riboflavin (ppm)	A2 Riboflavin (ppm)	A3 Riboflavin (ppm)
0.169	0.187	0.157
0.170	0.174	0.121
0.130	0.192	0.127
0.168	0.103	0.109
0.183	0.119	0.166
0.160	0.117	0.173
0.167	0.171	0.190
0.166	0.192	0.148
0.139	0.175	0.130
0.199	0.125	0.135
0.110	0.131	0.148

☒ Use first row as variable names

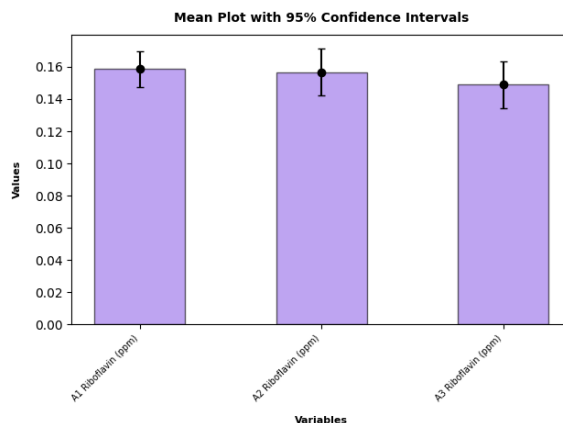
Calculate
Clean Data & Results

In this example, the first row contains the variable names, so the 'Use first row as variable names' checkbox must be selected. Then click 'Calculate' to process the data. To remove incorrectly entered values and clear the current output, click 'Clean Data & Results.'

For the 'Bulk Data' option, the results include a table of commonly used descriptive statistics and a chart showing the mean of each variable with its 95% confidence interval, as illustrated below.

Results

Variable	Size	Min.	Max.	Range	First Quartile (25%)	Median	Third Quartile (75%)	Mean	Sample Variance	Sample Std. Dev.	Std. Error	Margin of Error (95% conf.)	C.I. of Mean (95% conf.)	Skewness	Kurtosis	Shapiro-Wilk W	Shapiro-Wilk p
A1 Riboflavin (ppm)	16	0.11	0.199	0.089	0.14425	0.1665	0.16925	0.15875	0.00051593	0.022714	0.0056785	± 0.01113	0.14762 ; 0.16988	-0.4259	0.1106	0.9628	0.7121
A2 Riboflavin (ppm)	16	0.103	0.192	0.089	0.1295	0.1655	0.17875	0.15669	0.00089983	0.029997	0.0074993	± 0.014699	0.14199 ; 0.17139	-0.4173	-1.266	0.9102	0.1174
A3 Riboflavin (ppm)	16	0.106	0.199	0.093	0.1255	0.148	0.17225	0.149	0.00087813	0.029633	0.0074083	± 0.01452	0.13448 ; 0.16352	0.1654	-1.205	0.9537	0.5511

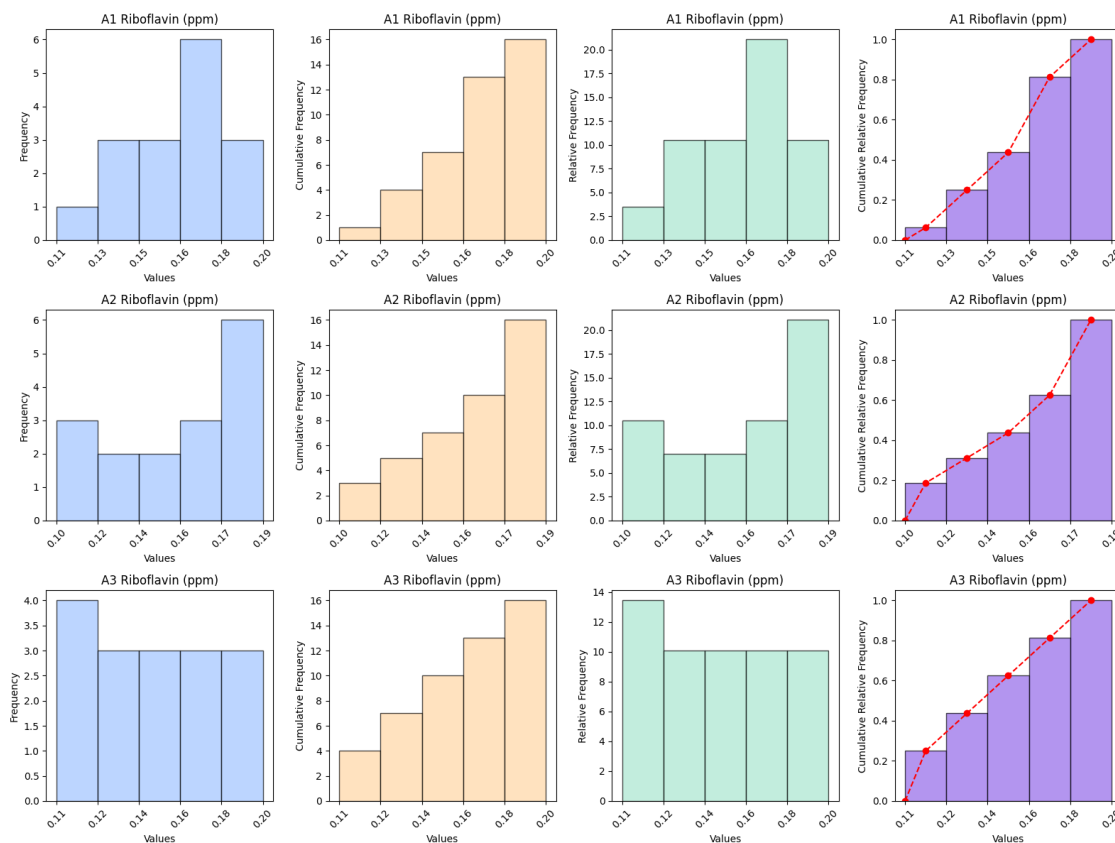


The variables do not need to contain the same number of observations. Missing values are excluded from the calculations, so the effective sample size is determined separately for each variable.

After a calculation is performed, the system retains the entered data. When another option, such as 'Histograms,' is selected, the same data will already be available and can be analyzed by clicking 'Calculate.'

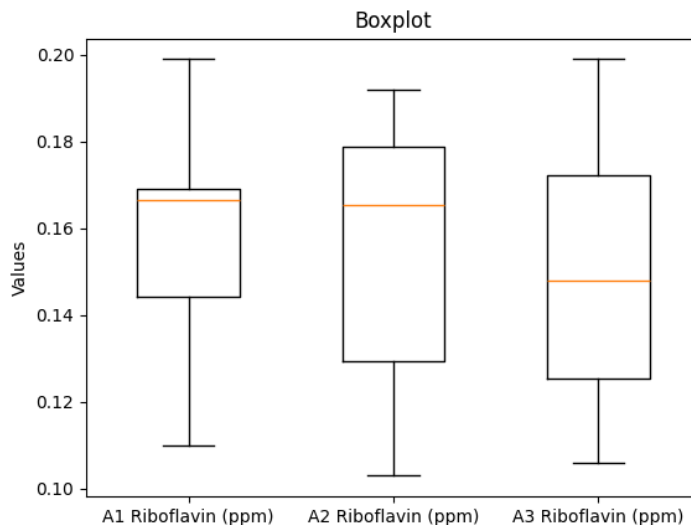
Click 'Calculate' to display the results for the selected option. The existing data may also be cleared and replaced with a new dataset from any of the available options.

The histogram results include frequency histograms, relative frequency histograms, cumulative frequency histograms, and cumulative relative frequency histograms with frequency polygons.





The procedure for the final option, 'Boxplot,' is similar. After selecting the option, click 'Calculate' to generate boxplots for the entered variables.



3. Probability Functions and Tables.

The Probability Functions and Tables section provides tools for working with four commonly used probability distributions:

- Standard Normal distribution
- Student's t distribution
- Chi-Square distribution
- Fisher-Snedecor F distribution

To access this section, click Probability Functions in the sidebar. The top navigation menu contains the Probability Functions calculator together with the corresponding statistical tables for each distribution.

The statistical tables display the relationship between values of the selected random variable and their associated cumulative probabilities. Users may consult these tables directly or use the Probability Functions calculator to obtain the same values interactively.

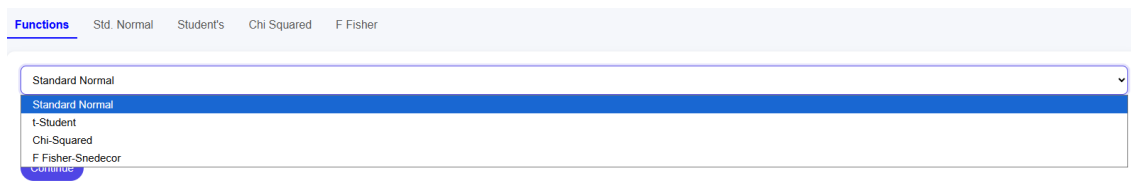
The calculator can be used in two directions:

- enter a value of the random variable to calculate its associated probability;
- enter a probability to calculate the corresponding value of the random variable.

Depending on the selected distribution, the calculator may also require one or more degrees of freedom.

3.1 Using the Probability Functions Calculator.

Select Functions from the top menu. A panel will appear with a drop-down list containing the available distributions.



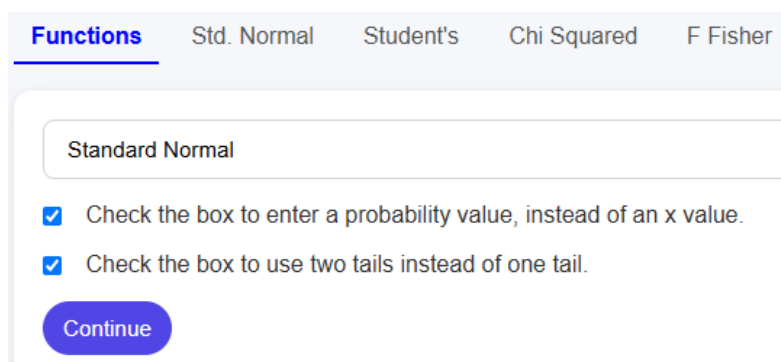
After selecting a distribution, the input fields and available options will change according to the parameters required for that distribution.

For the Standard Normal distribution, users can choose between the following calculation types:

- calculate a probability from a specified value of z ;
- calculate a value of z from a specified probability;
- calculate results for a single value;
- calculate results for an interval defined by two values.

The first checkbox changes the direction of the calculation. When it is left unselected, the user enters a value of the random variable and the platform calculates the corresponding probability. When it is selected, the user enters a probability and the platform calculates the corresponding value of the random variable.

The second checkbox enables interval calculations. When it is left unselected, the calculation is performed for a single value. When it is selected, two values must be entered to define the interval.



After selecting the required options, click Continue. The platform will then display the input fields needed for the selected calculation. Enter the requested values and click Submit to generate the numerical results and the corresponding probability distribution graph.

3.2 Standard Normal Distribution.



The Standard Normal distribution is a continuous probability distribution with mean equal to zero and standard deviation equal to one. It is one of the most widely used distributions in statistics because many statistical methods are based on its properties or can be transformed into this scale.

The Probability Functions calculator allows users to perform both probability calculations and inverse probability calculations using the Standard Normal distribution. In addition to the numerical results, the platform generates a graph highlighting the corresponding area under the probability density curve, making the interpretation of the results more intuitive.

3.2.1 Calculating a Probability from a z-Value.

To calculate a probability from a given value of the standardized variable (z):

1. Select Standard Normal Distribution from the distribution list.
2. Leave the Calculate value from probability option unchecked.
3. Leave the Interval calculation option unchecked if the calculation involves a single value.
4. Click Continue.
5. Enter the desired z-value.
6. Click Submit.

The platform displays:

- the cumulative probability associated with the specified z-value;
- the complementary probability;
- a graphical representation of the Standard Normal distribution with the corresponding area highlighted.

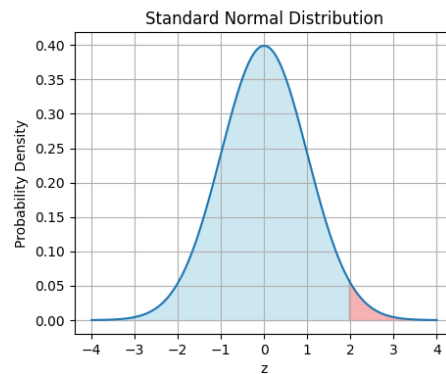
The screenshot shows a web interface for statistical functions. At the top, there are tabs: 'Functions', 'Std. Normal', 'Student's', 'Chi Squared', and 'F Fisher'. The 'Functions' tab is active. Below the tabs, there is a section for 'Standard Normal'. It contains two checkboxes: 'Check the box to enter a probability value, instead of an x value.' and 'Check the box to use two tails instead of one tail.' Both are unchecked. Below these is a 'Continue' button. Underneath, it says 'Enter a value for z:' followed by a text input field containing '1.96'. At the bottom of this section is a 'Submit' button.

For example, entering $z = 1.96$ produces the cumulative probability

$$P(Z \leq 1.96) = 0.9750$$

meaning that approximately 97.5% of the observations in a Standard Normal distribution are expected to be less than or equal to 1.96.

For a z value of: 1.96
Probability to the left (blue): 0.97500
Probability to the right (red): 0.02500



3.2.2 Calculating the Probability for an interval.

Many statistical applications require calculating the probability that a random variable falls between two values.

To perform this calculation:

1. Select Standard Normal Distribution.
2. Leave Calculate value from probability unchecked.
3. Check Interval calculation.
4. Click Continue.
5. Enter the lower and upper z-values.
6. Click Submit.

Functions

Std. Normal

Student's

Chi Squared

F Fisher

Standard Normal

☐ Check the box to enter a probability value, instead of an x value.

☒ Check the box to use two tails instead of one tail.

Continue

The platform calculates the probability contained within the selected interval and displays the corresponding shaded region on the graph.

For example, entering

- Lower value = -1.96
- Upper value = 1.96

produces

$$P(-1.96 \leq Z \leq 1.96) = 0.9500$$

indicating that approximately 95% of the observations are expected to lie within this interval.



Functions Std. Normal Student's Chi Squared F Fisher

Standard Normal

☐ Check the box to enter a probability value, instead of an x value.

☒ Check the box to use two tails instead of one tail.

Continue

Enter the first value for z:

-1.96

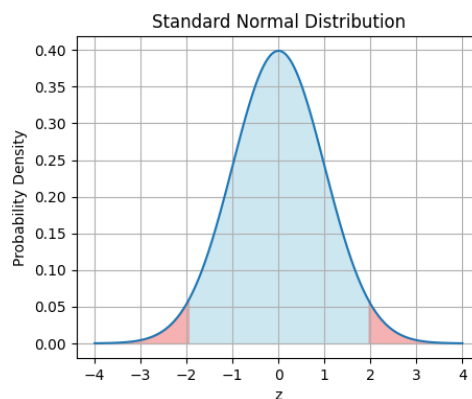
Enter the second value for z:

1.96

Submit

This interval is frequently used for constructing 95% confidence intervals and for performing two-sided hypothesis tests.

For the values of z: -1.96 and 1.96
Probability to the left of -1.96 (red): 0.02500
Probability between -1.96 and 1.96 (blue): 0.95000
Probability to the right of 1.96 (red): 0.02500



3.2.3 Calculating a z-Value from a Probability.

The calculator can also perform the inverse calculation by determining the z-value corresponding to a specified cumulative probability.

To perform this calculation:

1. Select Standard Normal Distribution.
2. Check Calculate value from probability.
3. Leave Interval calculation unchecked.
4. Click Continue.
5. Enter the cumulative probability.
6. Click Submit.

The platform returns the corresponding standardized value together with a graphical representation of the distribution.



Functions Std. Normal Student's Chi Squared F Fisher

Standard Normal

☒ Check the box to enter a probability value, instead of an x value.
☐ Check the box to use two tails instead of one tail.

Continue

Enter a value for p:

0.975

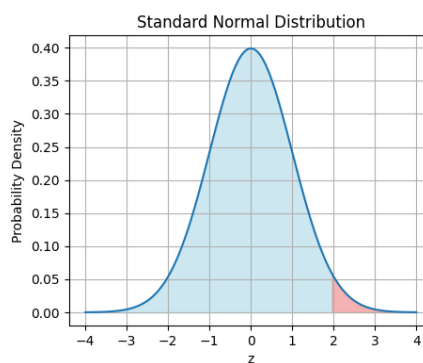
Submit

For example, entering a cumulative probability of 0.975 produces

$$z=1.96$$

which is one of the most frequently used critical values in inferential statistics.

**For a probability value of: 0.975 (blue)
Value of z: 1.95996**



3.2.4 Graphical Interpretation.

For every calculation, the platform automatically generates a graph of the Standard Normal distribution.

Depending on the selected calculation, the graph highlights:

- the cumulative probability to the left of a specified z-value;
- the probability contained within a selected interval;
- or the cumulative probability associated with a calculated critical value.

The graphical representation helps users interpret the numerical results and better understand the relationship between standardized values and probabilities.

3.3 Student's *t* Distribution.



The Student's t distribution is a continuous probability distribution that is widely used when making statistical inferences about a population mean, particularly when the population standard deviation is unknown and the sample size is small.

Unlike the Standard Normal distribution, the shape of the Student's t distribution depends on the degrees of freedom (df). As the degrees of freedom increase, the distribution becomes increasingly similar to the Standard Normal distribution.

The Probability Functions calculator allows users to calculate probabilities associated with t-values or to determine critical t-values from specified probabilities.

3.3.1 Calculating a Probability from a t-Value.

To calculate a cumulative probability from a specified t-value:

1. Select Student's t Distribution from the distribution list.
2. Leave Calculate value from probability unchecked.
3. Leave Interval calculation unchecked if the calculation involves a single value.
4. Click Continue.
5. Enter the required degrees of freedom.
6. Enter the desired t-value.
7. Click Submit.

Functions Std. Normal Student's Chi Squared F Fisher

t-Student

☐ Check the box to enter a probability value, instead of an x value.

☐ Check the box to use two tails instead of one tail.

Continue

Enter a value for t:

1.96

Degrees of freedom (positive integer):

10

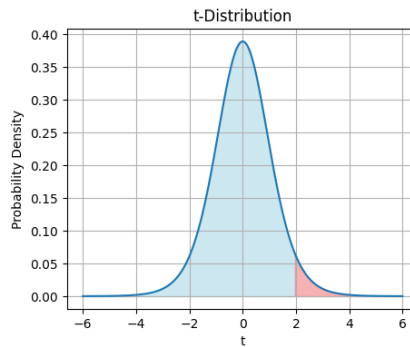
Submit

The platform displays:

- the cumulative probability;
- the complementary probability;
- a graph of the Student's t distribution with the corresponding probability region highlighted.

Because the Student's t distribution depends on the degrees of freedom, changing this parameter automatically updates both the numerical results and the graphical representation.

For a t value of: 1.96 with 10 degrees of freedom
Probability to the left (blue): 0.9608
Probability to the right (red): 0.0392



3.3.2 Calculating the Probability for an Interval.

The platform also allows users to calculate the probability that a random variable follows the Student's t distribution and falls within a specified interval.

To perform this calculation:

1. Select Student's t Distribution.
2. Leave Calculate value from probability unchecked.
3. Check Interval calculation.
4. Click Continue.
5. Enter the lower and upper t-values.
6. Enter the degrees of freedom.
7. Click Submit.

t-Student

☐ Check the box to enter a probability value, instead of an x value.

☒ Check the box to use two tails instead of one tail.

Continue

Enter the first value for t:

-1.96

Enter the second value for t:

1.96

Degrees of freedom (positive integer):

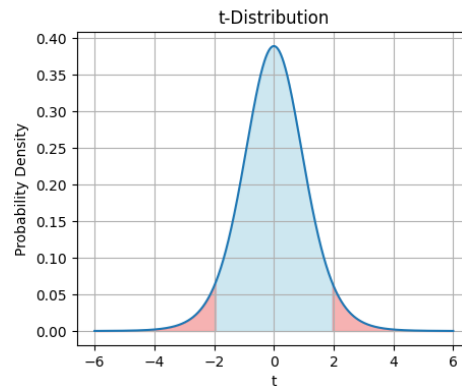
10

Submit

The platform calculates the probability contained within the selected interval and displays the shaded area under the probability density curve.

Interval probabilities are frequently used when interpreting confidence intervals and hypothesis tests based on the Student's t distribution.

For the values of t: -1.96 and 1.96 with 10 degrees of freedom
Probability to the left of -1.96 (red): 0.0392
Probability between -1.96 and 1.96 (blue): 0.9216
Probability to the right of 1.96 (red): 0.0392



3.3.3 Calculating a Critical t-Value from a Probability.

The inverse calculation determines the t-value associated with a specified cumulative probability.

To perform this calculation:

1. Select Student's t Distribution.
2. Check Calculate value from probability.
3. Leave Interval calculation unchecked.
4. Click Continue.
5. Enter the cumulative probability.
6. Enter the degrees of freedom.
7. Click Submit.

t-Student

☒ Check the box to enter a probability value, instead of an x value.
☐ Check the box to use two tails instead of one tail.

Continue

Enter a value for p:

0.95

Degrees of freedom (positive integer):

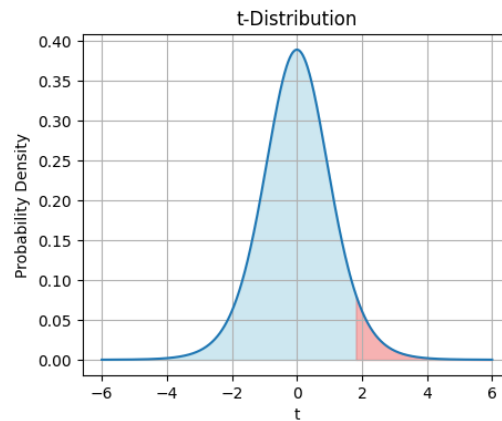
10

Submit

The platform returns the corresponding critical t-value and displays its position on the probability density curve.

Critical t-values are commonly used when constructing confidence intervals and performing one-sample, paired-sample, and independent-sample t-tests.

**For a probability value of: 0.95 (blue) with 10 degrees of freedom
Value of t: 1.8125**



3.3.4 Graphical Interpretation.

For every calculation, the platform generates a graph of the selected Student's t distribution using the specified degrees of freedom.

The graph highlights:

- the cumulative probability for a given t -value;
- the probability contained within an interval;
- or the location of a calculated critical value.

This graphical representation helps users understand how the degrees of freedom influence the shape of the distribution and the associated probabilities.

3.4 Chi-Square Distribution.

The Chi-Square distribution is commonly used in statistical inference for procedures involving population variances, goodness-of-fit tests, and contingency table analyses.

The operation of the Probability Functions calculator is identical to that described for the Standard Normal and Student's t distributions. The user selects the Chi-Square distribution, specifies the required degrees of freedom, chooses whether to calculate a probability or a critical value, and then enters the requested values after pressing Continue.

As with the previous distributions, the platform displays both the numerical results and a graphical representation of the corresponding probability density function.

3.5 Fisher-Snedecor F Distribution.

The F distribution is primarily used in procedures that compare variances, including analysis of variance (ANOVA), regression analysis, and variance ratio tests.



The calculation procedure is the same as that described for the previous probability distributions. The only difference is that the F distribution requires two degrees of freedom, corresponding to the numerator and denominator.

After selecting the desired calculation type, entering the required parameters, and pressing Submit, the platform displays the calculated probabilities or critical values together with the corresponding probability distribution graph.

4. Inference.

The Inference module provides statistical procedures for drawing conclusions about populations based on sample data. These tools allow users to estimate unknown population parameters, evaluate statistical hypotheses, assess whether data follow a specified probability distribution, and perform non-parametric analyses when the assumptions required for parametric methods are not satisfied.

To access this module, click Inference in the sidebar. The top navigation menu is organized into four groups of statistical procedures:

- Confidence Intervals
- Hypothesis Testing
- Distribution Testing
- Non-Parametric Testing

Each group contains several statistical methods designed for different analytical situations. Some procedures require summary statistics, whereas others use raw data entered directly into the application.

Although the required input varies among procedures, the overall workflow throughout the Inference module is consistent. Users first select the desired statistical procedure, enter the requested information, specify the confidence level when applicable, and then click Calculate to generate the results.

Most procedures produce both numerical and graphical outputs. The numerical results summarize the statistical calculations, while the graphs facilitate interpretation and provide a visual representation of the analysis.

4.1 Confidence Intervals.

The Confidence Intervals section is used to estimate unknown population parameters together with a measure of the uncertainty associated with those estimates. Rather than providing a single value, confidence intervals define a range of plausible values for the parameter of interest at a specified confidence level.

The platform includes confidence interval procedures for:

- Mean
- Variance
- Proportion
- Difference of Means
- Ratio of Variances



- Difference of Proportions
- Paired Mean Difference

The appropriate procedure depends on the parameter being estimated and on the available information, such as whether the population variance is known, whether two independent samples are being compared, or whether the data correspond to paired observations.

4.1.1 General Workflow.

Regardless of the selected procedure, all confidence interval calculations follow the same general workflow.

After selecting a confidence interval type, the corresponding input panel is displayed. Depending on the selected procedure, the application accepts either summary statistics or raw observations.

When summary statistics are required, the information is entered by copying and pasting values from a spreadsheet. Depending on the selected procedure, the required information may include sample size, sample mean, sample variance, population variance, sample proportion, or other summary statistics.

For procedures based on raw data, each variable should be entered in a separate column. Variable names may optionally be included in the first row by selecting the Use first row as variable names option.

The desired confidence level should then be specified. Unless indicated otherwise, the platform uses a default confidence level of 95% when no value is entered. The application also informs the user whenever a default value has been applied.

After clicking Calculate, the platform generates the numerical results together with graphical representations that facilitate interpretation. The entered data remain available while navigating within the module and can be removed at any time by clicking Clear Data & Results.

Example datasets are available on the Web of Statistics website for each confidence interval procedure. These examples illustrate the required data structure and show how the values should be organized before being copied and pasted into the platform.

4.1.2 Confidence Interval for the Mean.

The Confidence Interval for the Mean estimates the range within which the true population mean is expected to lie for a specified confidence level.

The platform provides two alternative procedures depending on whether the population variance is known or unknown. When the population variance is known, the interval is calculated using the Standard Normal distribution. Otherwise, the Student's t distribution is used together with the sample variance.

Depending on the selected procedure, the required input includes the sample size, sample mean, population or sample variance, and the desired confidence level. Multiple variables may be analyzed simultaneously by entering each variable in a separate column.

The following table presents two examples of the required data format for calculating confidence intervals for the mean under both available cases: known and unknown population variance. In each example, two variables are analyzed, although a single variable or several variables may be entered. The shaded cells



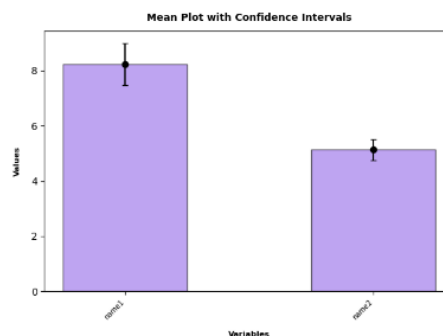
indicate the data that should be copied and pasted into the platform. The first row, containing the variable names, is optional.

Known Population Variance (sigma sqrd)			Unknown Population Variance (sigma sqrd)		
Var names (optional)	name1	name2	Var names (optional)	name1	name2
number of elements (n)	50	50	number of elements (n)	15	100
sample mean (X_bar)	4.63	30.5	sample mean (X_bar)	8.23	5.13
population variance (sigma sqrd)	0.2916	5.5	sample variance (S sqrd)	2.79	3.6
Confidence Level (%)	99	99	Confidence Level (%)	90	95

The following figure shows the results obtained for the confidence intervals of the two variables (name1 and name2) in the case of unknown population variance, using the data shown in the shaded cells on the right-hand side of the previous table.

Results

Variable	number of elements	Sample Variance	Confidence Level	Mean	Confidence Interval
name1	15.0	2.79	90.0	8.23	(7.47039, 8.98961)
name2	100.0	3.6	95.0	5.13	(4.75352, 5.50648)



The results include the estimated confidence interval together with a graphical representation that displays the point estimate and its corresponding confidence limits, facilitating comparison among multiple variables.

4.1.3 Confidence Interval for the Variance.

This procedure estimates the confidence interval for a population variance using the Chi-Square distribution.

Two alternative procedures are available depending on whether the population mean is known. When the population mean is unknown, the interval is calculated from the sample variance. When the population mean is known, the variance computed with respect to the population mean must be provided.

The required input includes the sample size, the corresponding variance estimate, and the desired confidence level.



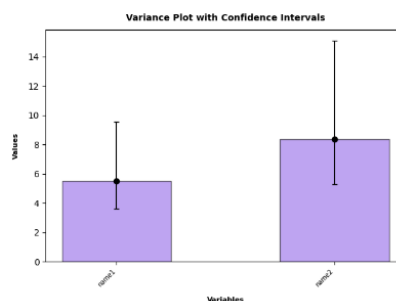
The following table presents two examples of the required data format for calculating confidence intervals for the variance under both available cases: known and unknown population mean. In each example, two variables are analyzed, although a single variable or several variables may be entered. The shaded cells indicate the data that should be copied and pasted into the platform. The first row, containing the variable names, is optional.

Known Population mean (μ)			Unknown Population mean (μ)		
Var names (optional)	name1	name2	Var names (optional)	name1	name2
number of elements (n)	20	40	number of elements (n)	25	30
sample variance (S sqrd)	0.8	1.5	sample variance (S sqrd using population mean)	5.5	8.35
Confidence Level (%)	95	99	Confidence Level (%)	90	95

The results present the lower and upper confidence limits together with a graphical representation of the estimated interval. The following figure shows the results obtained for the confidence intervals of the two variables (name1 and name2) in the case of unknown population mean, using the data shown in the shaded cells on the right-hand side of the previous table.

Results

Variable	number of elements	Confidence Level	Sample Variance	Confidence Interval
name1	25.0	90.0	5.5	(3.6249, 9.5318)
name2	30.0	95.0	8.35	(5.2961, 15.09)



4.1.4 Confidence Interval for a Proportion.

This procedure estimates the confidence interval for a population proportion.

The required information consists of the observed sample proportion together with the sample size and the desired confidence level. Since proportions represent probabilities, all entered values must lie between 0 and 1.

The platform calculates the confidence interval for each entered sample and presents both the numerical limits and a graphical representation of the estimated interval.



4.1.5 Confidence Interval for the Difference of Means.

This procedure estimates the confidence interval for the difference between the means of two populations.

Four alternative cases are available:

- Known population variances
- Unknown but equal population variances
- Unknown and unequal population variances
- Paired samples

The first three procedures require summary statistics for the two samples, whereas the paired-sample procedure requires the original observations because the calculations are performed using the differences between paired measurements.

The results include the estimated confidence interval together with a graphical representation. When the interval contains zero, there is insufficient evidence to conclude that the population means differ at the selected confidence level.

4.1.6 Confidence Interval for the Ratio of Variances.

The confidence interval for the ratio of two population variances is based on the Fisher–Snedecor F distribution.

The required input consists of the sample sizes and sample variances for two independent samples. The desired confidence level may also be specified.

The graphical output includes the estimated ratio together with its confidence interval. A reference line at 1 is displayed because a ratio equal to one indicates equal population variances. When the confidence interval includes this value, the observed difference between variances is not statistically significant at the selected confidence level.

4.1.7 Confidence Interval for the Difference of Proportions.

This procedure estimates the difference between two population proportions.

The required information consists of the sample sizes and the observed proportions for two independent samples. The desired confidence level can also be specified.

The resulting graph includes a reference line at 0, representing the absence of a difference between the two proportions. When the confidence interval contains zero, the observed proportions cannot be considered significantly different at the selected confidence level.

4.1.8 Interpretation of the Results.

All confidence interval procedures available in Web of Statistics produce numerical and graphical results.



The numerical output reports the estimated parameter together with its lower and upper confidence limits. The graphical output provides a visual representation of the interval, allowing users to compare estimates across variables or determine whether a reference value (such as 0 for differences or 1 for variance ratios) lies within the estimated interval.

Confidence intervals should always be interpreted in conjunction with the selected confidence level. Higher confidence levels produce wider intervals, reflecting greater uncertainty, whereas lower confidence levels result in narrower intervals but provide less confidence that the interval contains the true population parameter.

4.2 Hypothesis Testing

The Hypothesis Testing section provides procedures for evaluating statistical claims about population parameters using sample data.

The available tests closely correspond to the parameters considered in the Confidence Intervals section. Hypothesis tests can be performed for:

- Mean
- Variance
- Proportion
- Difference of Means
- Ratio of Variances
- Difference of Proportions

Depending on the selected parameter, additional subcases are available. For example, tests involving population means may depend on whether the population variance is known or unknown, while comparisons between two means may depend on assumptions regarding the population variances or whether the observations are paired.

4.2.1 General Workflow

The general workflow is similar across the different hypothesis tests.

After selecting the parameter and the appropriate subcase, users enter the required sample information and select the alternative hypothesis to be evaluated. Three types of tests are available:

Right-tailed test, when the alternative hypothesis proposes that the parameter is greater than the reference value.

Left-tailed test, when the alternative hypothesis proposes that the parameter is lower than the reference value.

Two-tailed test, when the alternative hypothesis proposes that the parameter differs from the reference value.

Depending on the selected procedure, the required input may include sample size, sample mean, sample variance, population variance, sample proportion, reference parameter values, and confidence level.

Multiple variables can be evaluated simultaneously when the selected procedure allows it. Each variable is entered in a separate column, and variable names may optionally be included in the first row.



After clicking Calculate, the platform reports the formulated hypotheses, test statistic, critical value or rejection criterion, p-value, and statistical decision, together with graphical representations when applicable.

Example datasets are available on the Web of Statistics website for each hypothesis-testing procedure. These examples illustrate the required data structure and show how the values should be organized before being copied and pasted into the platform.

4.2.2 Interpretation of Hypothesis Tests

The statistical decision is based on the selected significance level and the evidence provided by the sample.

When the p-value is lower than the significance level, the null hypothesis is rejected. Otherwise, there is insufficient statistical evidence to reject it.

Graphical outputs indicate the rejection region or regions and the position of the calculated test statistic, providing a visual interpretation of the decision.

For two-tailed tests, the significance level is distributed between both tails of the corresponding probability distribution.

This general interpretation applies to tests involving means, variances, proportions, differences between means or proportions, and variance ratios.

4.3 Distribution Testing

The Distribution Testing section provides methods for evaluating whether sample data are compatible with a specified probability distribution.

Two tests are available:

Kolmogorov-Smirnov Test
Shapiro-Wilk Test

The Kolmogorov-Smirnov test can be used to assess compatibility with the distributions available in the platform, including the normal and uniform distributions. The Shapiro-Wilk test is specifically intended to evaluate the assumption of normality.

Users may enter one or more variables as raw observations, with each variable placed in a separate column. Variable names may optionally be included in the first row.

For the Kolmogorov-Smirnov test, the distribution to be evaluated must also be selected.

After clicking Calculate, the platform provides the test statistic, p-value, and corresponding statistical decision for each variable.

When the p-value is lower than the selected significance level, the null hypothesis that the data are compatible with the specified distribution is rejected.

Example datasets available through the Web of Statistics website illustrate the required input format for these procedures.



4.4 Non-Parametric Testing

The Non-Parametric Testing section provides alternatives for comparing two samples when the assumptions required by the corresponding parametric methods are not appropriate.

Two procedures are available:

- Mann-Whitney-Wilcoxon Test, for two independent samples.

- Wilcoxon Signed-Rank Test, for paired or related samples.

In both cases, raw observations are entered in two columns. Variable names may optionally be included in the first row.

The Mann-Whitney-Wilcoxon test is appropriate when observations in the two groups are independent, whereas the Wilcoxon Signed-Rank test is designed for paired measurements, such as before-and-after observations collected from the same experimental units.

After clicking Calculate, the platform reports the corresponding test statistic, p-value, and statistical decision.

5. Statistical Process Control Charts

The Control Charts module provides tools for monitoring process behavior and identifying variations that may indicate changes in process stability.

The available methods include:

Shewhart charts for variables:

- X-bar and Range (X-R)
- X-bar and Standard Deviation (X-S)
- Median charts
- Individuals charts

Shewhart charts for attributes:

- p chart
- np chart
- c chart
- u chart

The module also includes:

- Pre-control charts
- EWMA charts
- CUSUM charts

The constants used for constructing Shewhart control charts are based on standard control-chart constants.

5.1 Shewhart Charts for Variables

X-R, X-S, and Median charts are based on measurements collected in rational subgroups. Numerical observations are copied and pasted into the data input area according to the required subgroup structure.



After calculation, the platform provides summary information together with the corresponding control charts, including the center line and control limits.

For an Individuals chart, observations are entered as a single series. Moving ranges between consecutive observations are used to estimate process variability.

5.2 Shewhart Charts for Attributes

Attribute control charts are designed for data based on counts of defective units or defects.

The required information depends on the selected chart:

p chart: proportion of defective units.

np chart: number of defective units.

c chart: number of defects when the opportunity for defects remains constant.

u chart: number of defects per unit when the number of inspected units may vary.

Depending on the procedure, one or two columns of numerical data are required.

5.3 Pre-Control Charts

Pre-control provides a simplified approach for evaluating process performance relative to predefined specification or tolerance limits.

In addition to the process observations, users must specify the nominal or target value and the applicable tolerance.

The resulting chart facilitates identification of observations located within the central, warning, or critical regions defined by those limits.

5.4 EWMA Charts

The Exponentially Weighted Moving Average (EWMA) chart is designed to detect relatively small and persistent changes in the process mean by assigning greater weight to recent observations.

The required parameters include the smoothing constant λ and a reference value for the process mean.

When no reference mean is provided, the platform uses the overall mean of the entered observations and informs the user that this default value has been applied.

5.5 CUSUM Charts

The Cumulative Sum (CUSUM) chart is another method designed to detect relatively small shifts in the process mean.



In addition to the observations, the procedure requires the parameters K and H and allows the user to specify a reference mean.

If no reference mean is entered, the overall mean of the dataset is used by default.

Example datasets for the different control charts are available on the Web of Statistics website and illustrate the required data structure for each procedure.

6. Analysis of Variance (ANOVA)

The ANOVA module provides statistical procedures for comparing the means of multiple groups and evaluating the effects of one or more categorical factors on a numerical response variable.

The platform includes:

- One-Way ANOVA
- One-Way ANOVA with Blocks
- Two-Way ANOVA
- Non-parametric alternatives

Post-hoc multiple-comparison procedures and diagnostic tests are also available when applicable.

6.1 One-Way ANOVA

One-Way ANOVA evaluates whether the mean response differs among the levels of a single factor.

For an analysis without blocks, the input data consist of one column identifying the factor levels and another containing the numerical response variable. The first row may optionally contain the variable names.

Users can specify the confidence level and, when appropriate, select a post-hoc multiple-comparison procedure.

Available post-hoc procedures include:

- Tukey HSD
- Bonferroni
- Scheffé
- Dunnnett
- Fisher's LSD
- Holm-Bonferroni
- Duncan

After clicking Calculate, the platform provides the ANOVA table and, when selected, the corresponding post-hoc comparisons.

Diagnostic outputs include the Shapiro-Wilk test for assessing normality and Levene's test for evaluating homogeneity of variances.

Graphical outputs include:

- means with confidence intervals
- boxplots



Together, these results allow users to evaluate both the statistical significance of the factor and the assumptions underlying the ANOVA model.

6.1.1 One-Way ANOVA with Blocks

One-Way ANOVA with blocks extends the previous design by incorporating a blocking variable to account for an additional source of variability.

The input data therefore contain three main elements:

- block
- factor
- numerical response

Factor and block levels may be represented by either numerical or text labels.

The statistical output follows the same general structure as One-Way ANOVA, including the ANOVA table, diagnostic tests, graphical analyses, and post-hoc comparisons when applicable.

6.2 Two-Way ANOVA

Two-Way ANOVA evaluates the effects of two factors on a numerical response and allows the interaction between those factors to be investigated.

The input dataset contains three principal columns:

- first factor
- second factor
- numerical response variable

The platform reports the significance of both main effects and their interaction.

As in One-Way ANOVA, post-hoc comparisons may be selected where applicable, and assumption checks include the Shapiro-Wilk and Levene tests.

The graphical outputs are similar to those generated for One-Way ANOVA, with the addition of an interaction plot, which helps users determine whether the effect of one factor changes depending on the level of the other factor.

6.3 Non-Parametric Tests

When the assumptions required for One-Way ANOVA are not adequately satisfied, the platform provides non-parametric alternatives.

For designs without blocks, the Kruskal-Wallis test is available. This procedure provides an alternative for comparing multiple independent groups.

When blocks are included in the experimental design, the Friedman test can be used instead.



Post-hoc procedures are also available:

Dunn test following Kruskal-Wallis.

Nemenyi test following Friedman.

The outputs are intentionally simpler than those generated by the parametric ANOVA procedures, but they provide the principal statistical information required to determine whether differences among treatments or groups are statistically significant.

7. Regression

The Regression module provides methods for modeling the relationship between a numerical response variable and one or more explanatory variables.

Two main groups of procedures are available:

Linear Regression

Non-Linear Regression

Linear regression can be performed either without interaction terms or with interactions among explanatory variables.

7.1 Linear Regression

For linear regression, the response variable Y is entered in the first column, followed by one or more explanatory variables X in the subsequent columns.

When a single explanatory variable is entered, the platform fits a simple linear regression model. When multiple explanatory variables are provided, a multiple linear regression model is fitted automatically.

Variable names may optionally be included in the first row.

After clicking Calculate, the results include information about the fitted model and estimated regression coefficients, together with statistical measures used to evaluate model performance.

For simple linear regression, interactive scatter plots display the observations and fitted regression line, both with and without the corresponding confidence band.

When two explanatory variables are used, the fitted relationship can be represented as an interactive three-dimensional surface. These plots may be rotated by dragging the mouse, allowing users to inspect the model from different perspectives.

Interactive standardized residual plots are also provided to support model diagnostics.

7.1.1 Models with Interaction Terms

When the interaction option is selected, the fitted linear model includes interaction terms between explanatory variables.

The required data structure is unchanged: the response variable is entered first, followed by the explanatory variables.



The output follows the same general structure as linear regression without interactions, including model coefficients, graphical representations of the fitted relationship, and standardized residual plots.

Example datasets illustrate the required format for simple regression, multiple regression, and models with interaction terms.

7.2 Non-Linear Regression

The Non-Linear Regression section fits predefined nonlinear functions to a numerical response variable Y and a single explanatory variable X.

The input therefore consists of two columns, with optional variable names in the first row.

After entering the data, users select the desired model from the available functions. These include:

- Quadratic
- Cubic
- Exponential
- Logarithmic
- Logistic
- Power
- Sinusoidal
- Gaussian
- Hyperbolic
- Sigmoidal

After clicking Calculate, the platform provides the fitted model and its estimated parameters.

Interactive scatter plots display the observed data together with the fitted nonlinear function, both with and without confidence bands when available.

8. Multivariate Analysis

The Multivariate module provides exploratory methods for analyzing datasets containing multiple variables simultaneously.

The principal methods available are:

- Principal Component Analysis (PCA)
- Hierarchical Cluster Analysis
- k-Means Cluster Analysis

8.1 Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms a set of potentially correlated variables into a new set of uncorrelated variables called principal components.

The components are ordered according to the amount of variability in the original dataset that they explain. PCA can therefore be used both to reduce dimensionality and to identify patterns, similarities among samples, and variables that contribute strongly to the observed variation.



The numerical variables are entered in columns and individual samples in rows.

Depending on the data structure supported by the platform, additional information may optionally include:

- sample names
- variable names
- a grouping factor used to distinguish observations in graphical outputs

When these identifiers are not supplied, the platform automatically assigns default labels.

After clicking Calculate, the platform generates several outputs, including:

- two-dimensional score plot
- interactive three-dimensional score plot
- scree plot
- loading plots
- explained variance values
- loading matrix
- score matrix

The score plots help identify similarities, clusters, and unusual observations, while the loading information helps determine which original variables contribute most strongly to each principal component.

8.2 Cluster Analysis

The Cluster Analysis section provides unsupervised methods for identifying groups of similar observations.

Two approaches are available:

Hierarchical Clustering

k-Means Clustering

For both procedures, numerical variables are entered in columns and samples in rows. Sample names may optionally be included to facilitate identification in the resulting graphs.

Hierarchical Clustering

Hierarchical clustering progressively groups observations according to their similarity and represents the resulting structure as a dendrogram.

The implementation provided by Web of Statistics uses the Ward method for constructing the hierarchical grouping.

k-Means Clustering

k-Means partitions the observations into a predefined number of clusters.

In addition to the dataset, users must specify the desired number of clusters.



The resulting silhouette plot provides a graphical assessment of the clustering structure. The platform also reports the average silhouette width, which can be used as an overall indicator of how well observations have been assigned to their respective clusters.

Final Considerations

Web of Statistics provides an accessible environment for statistical analysis, data visualization, and exploratory modeling. Its tools cover descriptive statistics, probability distributions, statistical inference, statistical process control, analysis of variance, regression, and multivariate methods, together with additional analytical capabilities available elsewhere in the platform.

A common data-entry approach is used throughout much of the application, allowing users to transfer data directly from spreadsheets and obtain numerical and graphical results without installing specialized statistical software.

This manual is intended as a practical guide to the operation of Web of Statistics. It explains the structure of the platform, the information required by each procedure, and the principal outputs generated by the available tools. It is not intended to replace textbooks, courses, or other specialized references on statistical theory. Users should have an appropriate understanding of the assumptions, limitations, and interpretation of the statistical methods they apply.

Example datasets available through the Web of Statistics website provide additional guidance on the data format required by each procedure and can also be used to explore the functionality of the platform.

Web of Statistics is an evolving open-source project. New analytical tools, improvements, and additional features may be incorporated over time in response to developments in statistical methodology and feedback from users.